

Adidas: Sustainability Strategy and the Road to Digital Product Passports

Kim Eun-hie (Oracle Korea Consultant)

Beginning in 2027, the European Union's **Digital Product Passport (DPP)** initiative will usher in a new era in which sustainability must be demonstrated through **data**, not simply declared through corporate commitments. In the past, companies could promote ESG initiatives and environmental policies at the organizational level. Going forward, however, they will be expected to prove—at the individual product level—that their products are safe for both people and the environment. This shift also reflects changing consumer expectations, particularly among younger generations, who increasingly favor brands that demonstrate commitments to waste reduction, environmental stewardship, and ethical labor practices.

Against this backdrop, Adidas offers a compelling example of how fashion companies can prepare for the DPP era. One of its most notable initiatives is the **Futurecraft.Loop** running shoe. Manufactured from a single material—100% recyclable thermoplastic polyurethane (TPU)—the shoe is designed without adhesives. Unlike conventional sneakers, which combine multiple materials and are difficult to recycle, Futurecraft.Loop can be ground down, melted, and remanufactured into an entirely new pair of shoes. It is, quite literally, a product designed with zero waste in mind. Similarly, Adidas' **UltraBOOST Parley**, made using plastic waste recovered from coastal areas, repurposes an average of 11 plastic bottles per pair.

Adidas' sustainability strategy extends well beyond product design. The company has made significant investments in IT systems capable of managing supply chain data across its global operations. Under its **Own the Game** strategy, announced in 2021, Adidas committed to investing more than **€1 billion** in digital systems by 2025 while hiring over **1,000 IT professionals**. This initiative represents more than an expansion of online sales—it reflects the company's transformation into a digitally driven retail business.

At the heart of this strategy is **TrusTrace**, a Swedish supply chain traceability platform. Designed to build trust in product authenticity and sustainability, TrusTrace integrates supply chain information from raw materials and yarn to fabric production, dyeing, and garment manufacturing. It also links environmental certifications—including **OEKO-TEX** and the **Global Recycled Standard (GRS)**—to purchase orders (POs), creating a unified record of compliance.

Most certification documents, which traditionally exist as PDFs or spreadsheets, are automatically classified using OCR and AI. The system extracts key information, categorizes document types, validates their contents, links them to existing enterprise systems, and verifies certificate authenticity in real time.

One of the platform's most valuable capabilities is verifying both certificate authenticity and quantity reconciliation. AI continuously audits whether the quantity of environmentally certified fabric entering the supply chain matches the quantity of finished products leaving it. Its **Certified Material Compliance (CMC)** module tracks and verifies in real time whether specific fabrics and materials are genuinely certified and environmentally compliant. In simple terms, it creates a data-backed record proving both the composition of a garment and the origin of its materials.

Within just four months of deployment, Adidas had used the system to track more than **one million transactions** spanning over **10,000 materials and product styles** across **8,000 facilities**. Today, it processes millions of data transactions every month. By comparison, manually reviewing five million certification documents would reportedly take a single person approximately **19 years**. In this context, AI-powered automation is no longer optional—it is essential.

Adidas' circular economy strategy is equally comprehensive. Its product development is built around three complementary principles:

- **Reuse** – maximizing the reuse of waste materials.
- **Circular Design** – designing products with future regeneration and recyclability in mind.
- **Regenerative Materials** – incorporating biodegradable materials and sustainably farmed resources.

The company also established ambitious sustainability targets, including using **100% recycled polyester by 2024** and ensuring that **90% of its products meet defined sustainability standards by 2025**.

As the Digital Product Passport becomes mandatory in 2027, companies will be required to disclose production information and environmental impact data with unprecedented transparency. In the years ahead, competitive advantage in the fashion industry is likely to depend not only on design excellence but also on the quality of supply chain data. Adidas' approach provides one of the most practical and instructive models for companies preparing for the DPP era.